ComputerLM

Persistent State and Asynchronous Reasoning for Fast Computer Use

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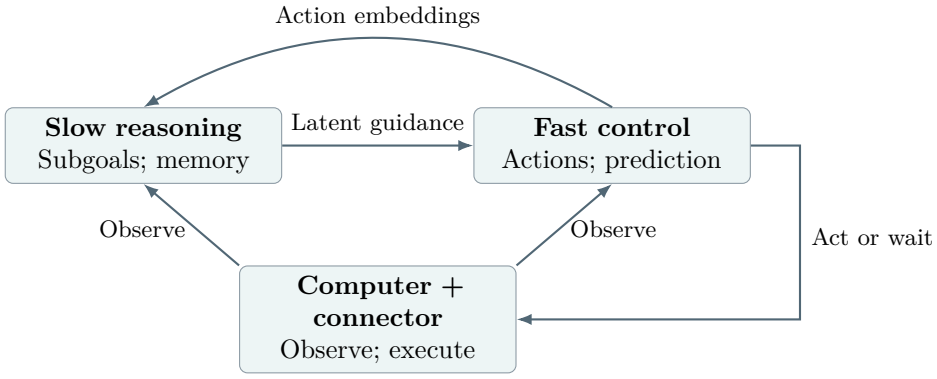
Summary. ComputerLM combines a small recurrent controller with a deeper reasoning path that continues while the controller acts. Both observe the interface and maintain living task state. Slow reasoning communicates subgoals through learned latent vectors; fast control grounds them into actions and predicts short-term consequences. The objective is to match the strongest available GUI systems on easy and medium browser/desktop workflows, prioritizing faster completion and then lower cost. The integrated model remains untrained; these gains are hypotheses to test.

1 Proposed architecture

Many workflows contain a planning decision followed by routine interactions. ComputerLM aims to reuse that decision, carry progress forward, and overlap useful reasoning with execution and application delays.



Fast path. A custom recurrent controller combines fresh visual features, action history, persistent state, and slow guidance. Only this path executes human-style GUI actions: clicking, typing, scrolling, dragging, keyboard shortcuts, or waits. The initial interface excludes shell, direct-file, DOM, and API shortcuts. Fast control predicts bounded action consequences; actual observations establish their effects.

Slow path. Adapt a suitably sized pretrained VLM into an input encoder, shared recurrent middle blocks, and guidance readout. It reasons continuously during an active task, including while fast control acts or waits. Guidance expresses dependencies such as *find X to obtain Y* in latent space. A trained interface makes those vectors useful to the controller; slow reasoning cannot directly act on the computer. The checkpoint and size remain open.

Perception and state. Begin with native VLM screen encoding plus available accessibility/UI text. Cache identical encoder inputs under matching settings; fully encode changed screens. Recurrent state survives interactions and pause/resume, then resets between unrelated tasks. Preserve exact instructions, text payloads, action acknowledgements, and evidence references alongside compressed state. Action embeddings report operations and arguments; they do not represent every possible environment state.

Expected savings. Reused decisions and maintained state aim to reduce repeated work; concurrency separately hides useful reasoning behind UI activity. Prediction helps only if preparation saves more work than it adds. Continuous slow reasoning can erase compute savings, so compare it with intermittent-slow control as well as a sequential baseline.

2 How the components work together

The **goal** specifies the desired result and constraints. The **state** tracks what is observed, pending, completed, or uncertain. An **action** uses both. These responsibilities do not require named latent dimensions or a fixed symbolic plan. Both paths retain access to the original instruction and exact values.

Representation	Owner	Purpose
Live UI belief	Fast path	Reconcile observations, issued actions, and pending effects.
Reasoning memory	Slow path	Retain intermediate understanding and unresolved dependencies.
Predicted futures	Fast path	Prepare bounded action-conditioned expectations, distinct from observed facts.
Subgoal guidance	Slow to fast	Communicate useful direction through the trained latent interface.
Exact records	Inputs/connector	Preserve text, execution status, and retrievable evidence.

Two forms of recurrence. Across interactions, learned state carries understanding into the next observation update. Within slow reasoning, shared blocks repeatedly refine a continuous representation without requiring written thinking tokens. The first aims to avoid rebuilding task understanding; the second spends computation on deliberation. State changes during a task while model weights remain fixed. Bounded recent context and access to older episode evidence complement the latent memory.

2.1 Example: attach a receipt and submit a report

1. Slow reasoning supplies the subgoal of attaching the receipt. Fast control grounds it into file selection and upload actions using current observations and the exact filename.

2. The connector reports that Upload was issued. State marks the attachment pending; a successful tool call does not establish that the report contains it.

3. While the application responds, fast prediction prepares likely next-state features. Slow reasoning continues through remaining requirements without waiting for a new action request.

4. A validation error updates the observed state and corrects the prepared future. Guidance depending on successful attachment is withdrawn; execution uses compatible direction or defers while slow reasoning revises it.

5. A later observation confirms the attachment. Progress advances, and fast control continues toward submission using current evidence.

Prediction is built into fast control. Before action selection, it can compare likely consequences of bounded candidates. After dispatch, it prepares the next decision while the UI changes. Predict compact features, not mandatory future screenshots. Actual observations correct predictions before dependent actions proceed; unbounded search is outside the initial scope.

Guidance has a lifetime. Associate guidance with source observations and action history, checking whether its assumptions still apply. Age alone is insufficient: older direction may remain useful, while fresh direction can conflict with new evidence. Withdraw an incompatible packet and publish a coherent replacement. A publication or executor wait does not stop slow reasoning; explicit pause, cancellation, or completion does. Compatibility checks and their training remain experimental.

3 Design rationale and development

Recurrent Memory Transformer carries learned memory between input segments [1]. Fast-in-Slow and DuoCore-FS provide robotics precedents for fast/slow cooperation and asynchronous latent guidance [2, 3]. In GUI research, iSHIFT conditionally requests more visual processing after latent deliberation [4]; GUI-JEPA studies lightweight latent action scoring [5]. CUWM studies action-conditioned UI prediction for action search [8]. These support components, not the complete architecture. Native VLM perception has a detailed precedent in Qwen2.5-VL [6].

Fresh evidence. Screens become available to both paths immediately. Compare next-cycle intake with trained early/middle/late adapters, preserving state and measuring reaction delay and recovery. Initial episodes follow one instruction; later live steering can revise goal state through the slow update interface and propagate revised guidance.

Adaptive execution. A central ablation compares immediate highest-scoring action selection with learned deferral for additional fast computation or fresher slow guidance. Prediction supports this decision; a prediction-disabled control tests its separate contribution. Both variants retain continuous slow reasoning and ordinary waits for pending UI operations. Action probability alone is not assumed to measure correctness.

Local constraints. Prefer Apple-Silicon-class inference with compact models, bounded state, and realistic unified-memory use. Initially target one GPU with CPU support. Custom scheduling may be needed; profile fit, slow progress, and fast tail latency. Exact hardware and serving software remain open. Approximate visual reuse is optional future work.

3.1 Staged training proposal

Training hardware and compute/data budgets remain unspecified. Begin with a small diagnostic:

1. **Preserve pretrained capability.** Establish one-pass parity and stable recurrent adaptation. Initially retain native perception; select frozen weights, adapters, and recurrent depth from development evidence.

2. **Learn execution and guidance together.** Train the controller and latent interface on chronological screen–action–outcome sequences. Include persistent memory, exact text access, failed actions, recovery, and delayed or missing guidance.

3. **Refine prediction and adaptive computation.** The fast predictor is part of the architecture from the outset. Supervise it with actual transitions and compare action/deferral policies. Collect learner failures for refinement, considering RL after useful supervised behaviour is established.

Training uses only information available at each decision. Keep held-out outputs and corrections outside training. Memory across observations and repeated internal reasoning depth are separate: the latter adds computation even when no thinking text is generated.

4 Evaluation objective

Use a balanced browser/desktop suite with fixed easy/medium criteria. Select the strongest available GUI references before testing; retain same-backbone sequential and component controls. Match observations, GUI actions, and resets. Equalize hardware for local controls; disclose hosted-reference hardware and network differences. Measure complete trajectories rather than action latency alone [7].

5 Evidence required and remaining decisions

Primary result: faster completion at matched accuracy. Predeclare the acceptable accuracy difference, task mix, repetitions, and improvement thresholds before held-out testing. Verify exact task outcomes independently and report failed attempts. Include timing on jointly solved tasks without dropping other attempts from the main results.

Secondary result: lower inference cost. Count both paths, perception, prediction, state updates, synchronization, verification, and recovery. Report task-time distributions, tail latency, memory, and compute separately. For local inference, measure energy where possible and disclose assumptions behind monetary estimates. A faster system can still consume more computation; claim joint speed/cost improvement only when both are measured.

Qualitative grading follows the repository contract: ChatGPT through the signed-in Codex CLI, after each episode. Judge latency is separate, judge failure stops the batch, and favourable qualitative scores cannot override a failed exact outcome check.

Open decisions. Text-composition responsibility and completion/stall reporting remain behaviour choices. Development experiments address:

- **Intake and memory:** cycle length, adapter location, state capacity, and exact-record access.
- **Learned interfaces:** guidance capacity, fusion, publication/compatibility checks, and whether action embeddings retain all required arguments.
- **Fast decisions:** controller capacity, prediction targets/horizons, and when deferral earns its time and compute.
- **Training and runtime:** data budget, trainable components, recurrent training windows, local hardware fit, and scheduling policy.

Contribution and limits. Seek evidence that the learned integration improves task speed while preserving success, with cost measured independently. The intended deliverable is trained weights, a recurrent/asynchronous runtime, and a computer connector. Stale guidance, lost constraints, prediction errors, and slow-path overhead are the main risks. Detailed decisions remain in `paper/design-questions.md`.

References

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